

# Smart E-Learning Assistant API

Reference documentation for the prediction and feedback-analysis service

Version 1.0

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## 1. Overview

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The API provides two independent features for the Smart E-Learning Assistant app:

- **Performance prediction** — given a student's demographic profile and their course activity so far, predicts whether they will pass or fail the module.
- **Feedback analysis** — given free-text student feedback, returns its sentiment and a short list of the issues it raises.

Both features are exposed as JSON HTTP endpoints.

## 2. Running the API

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From the `smart-elearning-api` directory:

```
pip install -r requirements.txt
cp .env.example .env
# edit .env and set GEMINI_API_KEY
```

```
uvicorn app.main:app --reload
```

The prediction endpoint requires trained model files in `models/` ( `pass_fail_pipeline.pkl` , `feature_stats.json` , `metadata.json` ). These are produced by running `student-performance-prediction.ipynb` . The feedback endpoint requires a valid `GEMINI_API_KEY` in `.env` .

Once running, interactive documentation is available at `http://127.0.0.1:8000/docs` .

### 3. POST /predict/performance

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#### POST /predict/performance

Predicts pass/fail risk from a student's demographic data and their course activity up to a given point in the course.

#### Why the request includes a course-progress value

The model is trained on data taken at several points through a course (10%, 25%, 40%, 55%, 70%, 85%, and 100% complete), not only on complete-course totals. This means it can be called mid-course without mistaking normal early-course activity for a risk signal. The request field `elapsed_fraction` tells the API how far into the course the student currently is; the service matches it to the nearest of the trained checkpoints above.

#### Request fields

| Field                | Type    | Required | Description                                                                                                                |
|----------------------|---------|----------|----------------------------------------------------------------------------------------------------------------------------|
| gender               | string  | yes      | "M" or "F"                                                                                                                 |
| highest_education    | string  | yes      | One of: "No Formal quals", "Lower Than A Level", "A Level or Equivalent", "HE Qualification", "Postgraduate Qualification" |
| imd_band             | string  | no       | Index of Multiple Deprivation band, e.g. "20-30%". Omit if unknown.                                                        |
| age_band             | string  | yes      | One of: "0-35", "35-55", "55<="                                                                                            |
| num_of_prev_attempts | integer | yes      | Number of times the student has previously attempted this module                                                           |
| studied_credits      | integer | yes      | Total credits the student is currently studying                                                                            |
| disability           | string  | yes      | "Y" or "N"                                                                                                                 |
| elapsed_fraction     | number  | yes      | How far through the course this request represents, from 0.0 (just started) to 1.0 (complete)                              |
| total_score          | number  | yes      | Sum of assessment scores achieved so far                                                                                   |
| avg_score            | number  | yes      | Average assessment score so far                                                                                            |

|                     |         |     |                                                                                               |
|---------------------|---------|-----|-----------------------------------------------------------------------------------------------|
| num_assessments     | integer | yes | Number of assessments submitted so far                                                        |
| total_clicks        | integer | yes | Total course-material clicks so far                                                           |
| recent_clicks       | integer | yes | Clicks in the last 14 days only. Must not exceed total_clicks.                                |
| date_registration   | integer | yes | Days between registration and the course start date (negative if registered before the start) |
| date_unregistration | integer | no  | Set only if the student has already unregistered; omit or send null otherwise                 |

## Example request

```
{
  "gender": "M",
  "highest_education": "A Level or Equivalent",
  "imd_band": "20-30%",
  "age_band": "0-35",
  "num_of_prev_attempts": 0,
  "studied_credits": 60,
  "disability": "N",
  "elapsed_fraction": 0.25,
  "total_score": 65,
  "avg_score": 65,
  "num_assessments": 1,
  "total_clicks": 120,
  "recent_clicks": 30,
  "date_registration": -20,
  "date_unregistration": null
}
```

## Response fields

| Field       | Type   | Description                                               |
|-------------|--------|-----------------------------------------------------------|
| prediction  | string | "Pass" or "Fail"                                          |
| probability | number | Model's estimated probability of passing, from 0.0 to 1.0 |
| risk_level  | string | "Low", "Medium", or "High", derived from probability      |

## Example response

```
{  
  "prediction": "Pass",  
  "probability": 0.82,  
  "risk_level": "Low"  
}
```

## 4. POST /feedback/analyze

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### POST /feedback/analyze

Analyzes free-text student feedback and returns its sentiment, a short list of issues it raises, and a one- or two-sentence summary. The feedback language is detected automatically (English, Arabic, or mixed), and the response is written in that same language.

### Request fields

| Field | Type   | Required | Description                             |
|-------|--------|----------|-----------------------------------------|
| text  | string | yes      | The feedback text, 1 to 4000 characters |

### Example request

```
{
  "text": "The recorded lectures are helpful but the assignment deadlines are too close together."
}
```

### Response fields

| Field     | Type             | Description                                                         |
|-----------|------------------|---------------------------------------------------------------------|
| sentiment | string           | "Positive", "Negative", or "Neutral"                                |
| issues    | array of strings | Short list of recurring problems or themes mentioned, empty if none |
| summary   | string           | One- or two-sentence summary of the feedback                        |

### Example response

```
{
  "sentiment": "Negative",
  "issues": ["assignment deadlines too close together"],
  "summary": "The student finds the recorded lectures useful but is frustrated by tightly clustered assignment deadlines."
}
```

## 5. GET /health

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GET /health

Reports whether the service is ready to serve real predictions.

### Response fields

| Field              | Type    | Description                                                                           |
|--------------------|---------|---------------------------------------------------------------------------------------|
| status             | string  | "ok" if both checks below pass, otherwise "degraded"                                  |
| model_loaded       | boolean | Whether the trained model files were found and loaded                                 |
| model_schema_valid | boolean | Whether the model's expected features still match what the service currently computes |
| gemini_configured  | boolean | Whether a Gemini API key was found at startup                                         |

## 6. Error responses

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| Status | When                                                                                                                                                       |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 422    | The request body fails validation — a missing field, a value outside its allowed range, an unrecognized option, or recent_clicks greater than total_clicks |
| 429    | Too many requests to /feedback/analyze from the same client in a short period                                                                              |
| 503    | /predict/performance was called but the model files are missing or invalid — check /health                                                                 |